

Aditya Punmiya

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EDUCATION

Manipal University Jaipur

Bachelor of Technology in Computer Science Engineering

CGPA: 8.57/10 [Dean's List]

Jaipur, Rajasthan, India

October 2021 - July 2025

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C#, Go, Bash

Frontend: React.js, React Router, Next.js, Tailwind CSS

Backend: FastAPI, Node.js with Express.js, Nest.js, gRPC, ASP.NET Core, Fiber, GraphQL, Deno, Bun.js, Socket.io

Databases: PostgreSQL, SQLite, MongoDB, Google Firebase, Redis

Tools & Technologies: Git, AWS, Azure, Digital Ocean, OCI, Vercel, Linux, Github Actions, Docker, Podman

WORK EXPERIENCE

Associate - Technology

Nav India

April 2026 - Present

- Architected the integration of a centralized password vault into the existing ecosystem, executing a seamless migration of legacy database credentials to enforce strict, decoupled secret management and mitigate security liabilities.

Trainee Associate - Technology

Nav India

July 2025 - April 2026

- Authored custom IronPython scripts to process in-memory financial data within a C# service, enabling complex calculations not available in the base application.
- Architected a custom, full-stack Role-Based Access Control (RBAC) framework from the ground up, implementing a secure authorization layer in FastAPI and modular, permission-aware React components to enforce granular access policies.
- Developed a high-throughput RabbitMQ consumer to orchestrate asynchronous data synchronization, utilizing idempotent logic for MongoDB inserts and updates to maintain real-time data consistency.
- Automated a dedicated database backup utility to perform daily snapshots of MongoDB instances, establishing a robust disaster recovery layer and ensuring long-term data integrity.
- Implemented a custom job-orchestration layer to monitor and cap simultaneous login sessions across distributed scrapers, leveraging MongoDB for real-time state management to prevent service-level bottlenecks.

Full Stack Intern

Blowbits Solutions LLP

Jan 2025 - June 2025

- Designed and developed a customer support ticketing system from scratch for both end-users and agents, building RESTful APIs using ASP.NET Core (.NET 8) and frontend components in React (TypeScript), with PostgreSQL for data storage and deployment on AWS.
- Built a complete attendance and leave management system using ASP.NET Core (.NET 8), React (TypeScript), PostgreSQL, and AWS, creating foundational HR functionality including API design, database schema, and responsive UI.
- Engineered a CAPTCHA recognition microservice using ASP.NET Core (.NET 8), ONNX Runtime, and Emgu.CV, integrating a pre-trained CNN model (converted to ONNX) for automated text extraction, with a focus on backend service design, image processing, and deployment on AWS.

Backend Intern

Techwondoe

September 2024 - January 2025

- Developed and maintained microservices using Nestjs, PostgreSQL, and TypeORM.
- Implemented GraphQL queries to fetch data from HubSpot CRM and integrated RESTful APIs to handle communication between the Next.js frontend and the Nestjs backend.
- Designed and built a notification microservice to handle notifications across the platform.
- Synced AWS Cognito users with Salesforce users, enabling users to log in through either service.
- Developed logic to generate and validate NCS-compliant UUIDv7 for Salesforce registrations, while AWS Cognito generated UUIDv7 for Cognito users, ensuring uniformity and proper validation across the system.

- Created logic to handle edge cases, including user-id conflicts and user registration flow across multiple authentication systems.

Full Stack Developer

Software Development Centre, Manipal University Jaipur

January 2024 - June 2024

- Developed the backend for my university's notesheet management platform using Node.js with Express.js and PostgreSQL. This project secured 1st place in the Software Development category under Minor Project at my university and is currently awaiting an Indian patent grant.

Research Intern

Language and Cognition Lab, IIT Kanpur

June 2023 - July 2023

- Implemented an algorithm for generating Hindi pseudowords into a web-based platform using Flask and Jinja, enabling practical experimentation.

PROJECTS

- **Distributed Microservice Platform**, Architected a strictly-isolated ecosystem powering a custom Next.js portfolio (Tenant #1), routing external traffic through a Go/Fiber API gateway to internal gRPC backend services (Auth, CMS Manager). Enforced zero-trust database security via restricted PostgreSQL roles and PL/pgSQL stored procedures, deploying on bare-metal via Podman. [Architecture](#)
- **Wallet**, Developed a personal expense tracking application using ASP.NET Core and Dapper ORM for high-efficiency data access, utilizing PL/pgSQL stored procedures for relational normalization; currently being refactored for ingestion into the centralized gRPC microservice ecosystem. [Wallet](#)
- **B21**, Engineered a Go-based stealth relay for high-security environments to retrieve data through restrictive firewalls via traffic camouflage, implementing TOTP authentication for zero-trust access and ensuring resilient data extraction. [GitHub](#)
- **Shell Aid**, Built a Go-based CLI tool using the Gemini AI API to transform natural language into cross-platform shell commands, featuring automated local environment/shell detection and safety-focused side-effect analysis. [GitHub](#)